

1.3 FBLA STUDENT ACTIVITIES

IST/CSP

Micro:bit Smart Hall Pass System

FBLA Event: Sales Presentation

Students design and build a Micro:bit-powered hall pass system intended to streamline student movement tracking and improve school operations. Students create a prototype, prepare a product pitch, and present the solution as a technology product.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback

The screenshot displays a Microsoft Teams assignment page for 'MJ2 - Hall Pass'. The top navigation bar shows the assignment title, due date (Sep 14 at 11:59 PM), and progress (46/46 Graded, 89.57 / 100 (90%) Average, 30/46 Students). The main content area is divided into two sections. The left section, titled 'Assignment', shows a code editor with a Micro:bit program. The program includes blocks for 'on button A pressed', 'radio send string', 'show string', 'repeat', and 'on radio received'. The right section, titled 'Assessment', shows a grade of 100% and a status of 'None'. Below the assessment, there are two assignment comments. The first comment is from 'Dina, Xia, Lauren, jacob' and the second is from 'Dina Madaoui'. Both comments praise the students' work and provide feedback on their presentation and the system they built.

MJ2 - Hall Pass
Due: Sep 14 at 11:59 PM - P4P6-Intro to Software
46/46 Graded 89.57 / 100 (90%) Average 30/46 Students
DM ✓ Dina Madaoui

Assessment
Grade (100 / 100) Status
100% None

Assignment Comments

DM Dina, Xia, Lauren, jacob
Dina Madaoui, Sep 14 at 8:35 PM

That was a great demonstration of team work and the ability to divide the activity of the teacher tool and the student tool to submit and accept Hall Passes,

since we experience difficulties in the beginning of the year with 'passes' we were able to demonstrate a system that could work and could be efficient to use in the classroom, Have you consider to create a prototype and build a professional solution with the microbit?

AP Computer Science Principles

What's the Odds? Powerball vs. Bitcoin Mining

FBLA Event: Computer Problem Solving

Students investigated and compared the probability of winning the Powerball lottery versus successfully mining Bitcoin. Students researched the problem, gathered supporting information, analyzed the mathematics required for comparison, explored probability concepts, and followed a design-thinking process to develop a proposed software solution before implementation.

Evidence Required

- One Graded Student Sample
- Teacher Feedback

The screenshot shows a presentation slide titled "Powerball research" with a list of bullet points and a mathematical formula. The sidebar on the right contains submission information.

Powerball research

- According to Powerball itself (powerball.com) Each ticket costs 2 dollars
- Doing the math, if you spent the whole 1000\$ on tickets, it would total to 500 tickets
- In order to play powerball, you'd pick 5 numbers from 1-69, as well as a Powerball number from 1-26
- Even though the amount of tickets may sound like we have good odds, the projected jackpot odds are very slim. Doing this math, this is the amount of possible combinations from 5 numbers between 1-69

$$\left(\frac{69}{5}\right) = \frac{69!}{5!(69-5)!} = 11,238,513$$

- Afterwards, we must multiply this by 26, which in total gives us 292,201,338, which means our final odds to win the jackpot prize in the powerball to 1 in 292201338. This is helped by our purchasing of 500 tickets, where it is lessened to 1 in 584,403.
- As well as this, we are almost guaranteed to be able to get more money from smaller prizes, but since it all falls on chance, we ultimately cannot bet on making a profit, let alone making it big

Submission to view: Aug 20 at 12:02 AM

Word Count: 577

Submitted Files: (click to load)
[Bitcoin Vs Powerball by Ismail2.pptx](#)

Assessment

Grade out of 100	Status
99	None

Assignment Comments

Great use of resources and also good analytical skills to identify the right probability model, including math equations to solve the two models before you are able to compare them and generate a solid conclusion of the odds.

AP Computer Science A

Technology Leadership Speaking Challenge

FBLA Event: Impromptu Speaking

Students receive a technology topic, are given twenty minutes to prepare, and deliver a five-to-seven-minute speech demonstrating communication, leadership, and technical understanding.

Evidence Required

- One Graded Student Sample

- Teacher Feedback

Game Design and Simulation

Environment and Emotional Design Analysis

Students evaluate how environmental design, audio, storytelling, color theory, level design, and gameplay mechanics affect player engagement and overall user experience.

Evidence Required

- Assignment Sheet
- One Graded Student Sample
- Teacher Feedback

PR-3 Environmental Game Design Review
Due: Aug 28 at 11:59 PM • PR-Game Design Animation and Sim-Torres

12/12 65.42 / 100 (65%) 9/12
Graded Average Students

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100% Late
Day 0.47
Due: Aug 28 at 11:59 PM

Mechanics

The player can move around the place, and interact with the robot citizens there. As you talk to the robots, they give you informations, clues, and even details about their daily life. This allow you to both find clues for your mission and immerse in the environment.

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Assignment Comments

i enjoyed it, thanks for sharing. x
Aldrin Torres, Aug 31 at 6:28 PM

As you went through observing and analyzing the gameplay you can realize the importance of the ambient, the environment, every detail counts for capturing the player, to help with the engagement with music, animations, objects, colors that the game intent to allow the player to follow the story through exploration. x

Rubric: **All completed**

Explain location
use of elements
Mechanics
Level of Intensity
How to spark Intensity
what creates emotions
thoughts about the overall story